# PRAMANEX TRACE: Evidence Is the Control Plane

A Reference Architecture and Operating Contract for Bounded System Assurance

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| DOCUMENT CLASS | OPERATING METHOD | CONTROL POSTURE |
|---|---|---|
| Source-grounded reference architecture | Track - Resolve - Assign - Control - Evaluate | Zero implicit trust / bounded release |

**Abstract**

Pharmaceutical decisions cross regulatory, quality, manufacturing, safety, laboratory and analytics systems. Their evidence often does not cross with them. A source is copied without its authority or version; an AI-generated candidate loses its retrieval and configuration context; a reviewer decision is recorded apart from the rationale that justified it. This fragmentation makes a plausible output difficult to reconstruct, correct or govern. TRACE specifies a vendor-independent evidence control plane that binds source authority, permission, version, transformation, candidate, validation, human disposition and operational outcome into one reviewable chain. Its operating method is Track, Resolve, Assign, Control and Evaluate. Every boundary has an explicit denial or quarantine state, including unknown authority, conflicting evidence, missing permission, unsupported material claims, stale approvals and unrecoverable execution. The architecture separates bounded AI preparation from professional decision rights and measures value only against a declared pilot baseline. It can be applied to multiple governed pharmaceutical workflow mandates without assuming that a single model, database or orchestration vendor supplies trust. TRACE is a reference architecture and operating contract. The supplied source establishes no production deployment, customer outcome, independent security assessment or organization-specific validation. Its central hypothesis is testable: preserving evidence and authority across the complete decision path can make AI-assisted work more reconstructable and correctable under controlled use.

**SOURCE-GROUNDED REFERENCE · Evidence and scope limits stated in the paper**

## 1. INTRODUCTION & THE EVIDENCE FRAGMENTATION CRISIS

The recurring systems failure is a broken handoff. A retrieved regulation may be current but inapplicable; an applicable passage may be translated without its original; a correct summary may never become an owned action. In each case, a downstream actor sees an answer but cannot reproduce the path to it.

Source/version fragmentation creates obsolete or duplicate authority. Requirement/impact fragmentation separates change description from affected work. AI execution fragmentation hides retrieval, tool and model state. Human-decision fragmentation leaves correction and rationale outside the record. Operations fragmentation converts retries and partial failures into apparent success.

TRACE treats a decision as a typed, version-bound object with a source lineage, context, accountable actor and terminal state. Its scope is evidence-to-action preparation within authorized workflows. An organization must define its intended use, source rights, roles, data retention, security boundary and acceptance cases before an operational pilot.

Time savings, evidence reuse, review-cycle improvement and reduced rework remain hypotheses until measured with an observation window and the cost of control included.

### 1.1 System Boundary

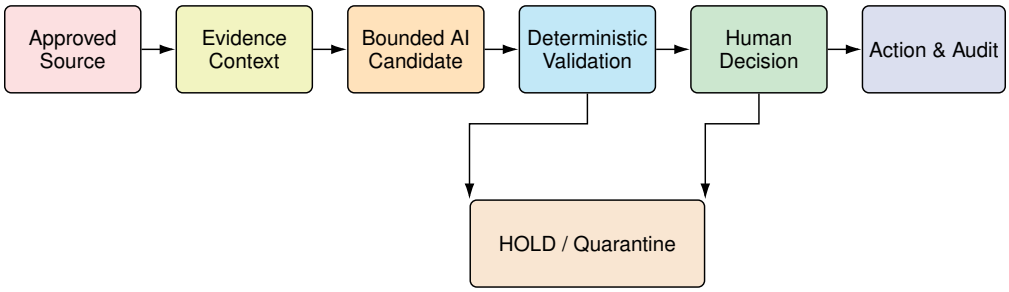
The control plane spans identity and permission; authoritative source capture; evidence context; bounded model or rule execution; deterministic validation; human disposition; and audit, recovery and outcome measurement. A model may propose a candidate, extraction, comparison or draft. It cannot grant its own source authority, accept residual regulated risk or manufacture a reviewer signature. Tool substitution is permissible only when equivalent evidence, authority and failure behavior are demonstrated.

### 1.2 Reviewable Record

For every material candidate, the record must answer: which approved source and version; which passage or object; which permission and purpose; which transformation and configuration; which conflicting or missing evidence; which named owner and reviewer; which revision was approved; and which downstream action was closed or held. The record is a graph of linked facts and decisions rather than a free-form transcript.

Reconstructability is a release property, not a reporting convenience.

**FIGURE 1 / EVIDENCE-TO-DECISION CONTROL PLANE**




Unknown authority, permission failure, contradictory evidence, stale decision, or failed critical control remain explicit recorded states.

Figure 1. Evidence-to-decision control plane. Proposed architecture; failures enter HOLD / quarantine.

| Stage | Control interpretation |
|---|---|
| Approved source + permission + version | Establish source authority, permitted purpose, retrieval time and version before content is trusted. |
| Exact evidence context | Preserve exact source coordinates, original-language truth, contradictions, lifecycle status and no-answer outcomes. |
| Bounded AI or rule candidate | Prepare typed candidates under a controlled configuration; do not grant authority or professional approval. |
| Deterministic validation | Check permission, grounding, lifecycle status, sensitive data, configuration, tool scope and output policy. |
| Human decision / HOLD | A named professional decides; source failure, permission failure, conflict or stale decision remains a first-class state. |
| Action, closure and audit | Execute scoped downstream action only after release; preserve closure, rollback and reconstructable lineage. |

This diagram is a proposed reference architecture. The candidate cannot bypass validation or replace the named professional disposition. A source failure, permission failure, contradictory evidence or stale decision remains a first-class recorded state.

## 2. THE T-R-A-C-E OPERATING PIPELINE

### 2.1 Track: authority before content

Track captures an approved source, retrieval time, source version, origin, permitted purpose and access decision. An observed document is not automatically authoritative. A first snapshot establishes a baseline, an inaccessible source records a source failure, and a stable snapshot records no change. These states must never be conflated. Unknown authority or missing permission blocks further release and preserves the attempted intake for investigation.

### 2.2 Resolve: bounded evidence, explicit uncertainty

Resolve links exact source coordinates to candidate requirements, changes or facts. It retains original-language truth, lifecycle status, contradictions and no-answer outcomes. Retrieval is permission filtered before ranking; a fluent synthesis cannot turn a conflicted source into one apparently settled claim. Unsupported material fields become evidence gaps or HOLD. The evidence packet retains original context and the transformation path so a reviewer can challenge an interpretation.

### 2.3 Assign: rights and accountability

Assign binds the candidate to an owner, a qualified reviewer, a purpose and a decision scope. Reviewer authority is checked at decision time and against the exact candidate revision. The orchestrator can route or request evidence but cannot substitute model confidence for professional authority. Unowned work is an explicit pending state. A corrected candidate creates a new revision; its prior review does not silently survive.

### 2.4 Control: deny unsafe transitions

Control applies deterministic checks to permissions, grounding, lifecycle status, tool scopes, sensitive data, configuration and output policy. A critical failure quarantines the candidate, prevents external side effects and records the reason. Recovery requires corrected evidence, a new validation run and, where material, fresh human review. A write action receives its own scoped authorization. Failed controls do not become exceptions merely because another aggregate score is high.

### 2.5 Evaluate: measure both utility and failure

Evaluate tests normal and adverse cases, disagreement, corrections, reliability, recovery and operational burden. The declared denominator and critical-case outcomes accompany any metric. Pilot value compares an observed baseline with review time, correction effort, evidence reuse and cost of operating controls. Incident, rollback and HOLD are outcomes in the evaluation, never missing rows. Expansion requires observed evidence and an accountable decision.

## 3. ZERO IMPLICIT TRUST: FORMAL AXIOMS

Let a be an actor, s a source, p a purpose, c a candidate, r a revision and d a decision. Authn(a) establishes identity only; Authority(s,p) establishes approved source fitness for a purpose; Permit(a,s,p) establishes authorized access; Grounded(c,s) establishes a checked evidential link; and Approve(a,c,r) is a scoped human disposition.

**Axiom 1**

```
Authn(a) != Authority(s,p)
```

Signing in proves neither that a source is official nor that its content applies to a case.

**Axiom 2**

```
NetworkLocation(s) != Permit(a,s,p)
```

A reachable endpoint, internal network or retrieved match does not confer a use right.

**Axiom 3**

```
Citation(c,s) != Grounded(c,s)
```

A citation string must resolve to the stored version and relevant location and survive context review.

**Axiom 4**

```
ModelLabel(m) != Fitness(m,p)
```

A model name or prior score cannot qualify a model for a changed intended use.

**Axiom 5**

```
Approve(a,c,r) => Authorized(a,p) AND
          Current(c,r) AND
       SufficientEvidence(c)
```

A stale or unauthorized review is rejected, with history retained.

**Release invariant**

```
Release(c,r) => Permit AND Authority
AND Grounded AND CriticalChecksPass
      AND HumanDecisionCurrent
```

Unknown, conflict, stale or failed values are not coerced to true.

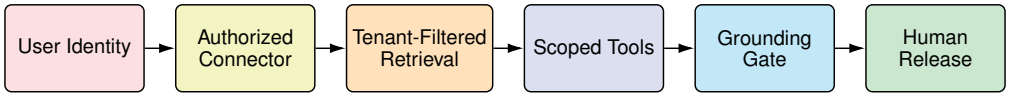
These are design invariants; implementation must prove them with positive and negative tests.

## 4. REFERENCE ARCHITECTURE & TRUST BOUNDARIES

The authority plane authenticates actors, determines purpose, enforces least privilege and records denials. The source plane stores approved originals, retrieval metadata, version and supersession; a content hash helps detect byte changes but cannot prove semantic correctness or completeness. The evidence plane creates permission-aware links between exact spans and candidate claims, preserving gaps and contradictory paths. The workflow plane manages idempotent transitions, queues, retries, waits and terminal failures. A bounded AI plane performs typed preparation under a controlled configuration. The assurance plane checks grounding, policy, safety and professional authority before a controlled output. The audit plane reconstructs every source-to-outcome dependency, including reversals and rollback.

The architecture is not a shared unrestricted knowledge pool. Each tenant, user, connector, retrieval operation, model call, tool and output has a separate trust boundary. A denied retrieval must produce no unauthorized semantic match. An output cannot launder data from a prohibited source through a model summary. Monitoring surfaces outages and latency while preserving the distinction between a system error and an authentic no-change observation.

**FIGURE 2 / TRUST BOUNDARIES AND DENIAL POINTS**




Each arrow is an independent authorization or evidence check; failure enters HOLD / quarantine. A permitted user cannot authorize an unapproved source; a valid source cannot cross a tenant boundary; a grounded candidate cannot self-approve.

Figure 2. Trust boundaries and denial points. Proposed architecture; failures enter HOLD / quarantine.

| Boundary | Independent control requirement |
|---|---|
| Identity / purpose | Authenticate the actor and bind intended purpose; identity alone does not prove source authority. |
| Connector / source | Authorize connector and source separately; reachable or internal does not equal permitted. |
| Tenant-filtered retrieval | Filter by tenant and permission before ranking so prohibited material never becomes a semantic match. |
| Scoped model and tools | Bind model configuration and tool permissions to intended use; write actions require explicit authorization. |
| Grounding + policy gate | Resolve exact evidence, conflicts and material gaps; deterministic checks govern release. |
| Human release | Verify reviewer authority and exact candidate revision; the system never self-approves. |

Every arrow is an independent authorization or evidence check. A permitted user cannot make an unapproved source authoritative; a valid source cannot cross a tenant boundary; and a well-grounded candidate cannot approve itself.

## 5. HUMAN AUTHORITY & WORKFLOW MANDATES

The orchestrator routes, holds and asks for evidence. A bounded AI component proposes structured candidates and uncertainty. A process reviewer corrects within delegated scope. A qualified regulatory, quality, medical, safety or supply professional makes the applicable professional disposition. An accountable owner accepts residual pilot risk. Every material output carries a named owner, evidence basis, revision, review state and release state.

TRACE's shared method can govern seven public mandates. These are mandate descriptions, not claims that seven products have been implemented. ClaimGraph-RD and Bharat-VPop are separate synthetic research artifacts; their reported benchmark results do not measure TRACE.

LEDGER Quality/compliance lineage.
BEACON Regulatory change triage.
LENS Promotional and medical review preparation.
AEGIS AI asset assurance.
ATLAS Permission-aware enterprise knowledge.
CANON Labeling comparison.
LIFELINE Supply/shortage signal triage.

## 6. VALIDATION, QUARANTINE & BOUNDED EXECUTION

First, freeze the intended use, users, source registry, exclusion list, material-field rules and qualified decision rights. Next, register versions of inputs, retrieval, model, workflow and tool policy. Execute normal, missing-source, contradictory-source, inaccessible-source, unauthorized-retrieval, stale-review and unsafe-tool cases against expected outcomes. Inspect both candidate quality and the control's refusal behavior.

When a safety gate fails, stop publication and side effects; record the failed control and exact execution state; quarantine the candidate and its dependent actions; alert the accountable owner;

request missing or corrected evidence; rerun the affected checks against a new revision; and obtain new human disposition where the change is material. Preserve the old revision for forensic replay. Critical permission or privacy failures trigger incident and rollback procedures under the deploying organization's policy.

An acceptance packet contains source and permission decisions, expected/actual test cases with denominators, reviewer rationale, adverse-case outcomes, security evidence, backup/recovery results and residual-risk approval. Release remains denied when critical evidence is missing. FDA computer software assurance guidance for medical-device production and quality management software can inform a risk-based assurance discussion; its specific scope cannot be generalized into a universal pharmaceutical certification.

### 6.1 ASSURANCE PACKET AND FAILED-GATE TRANSITION

**ASSURANCE PACKET - RELEASE PRECONDITION**

The intended-use contract identifies users, decision influence, exclusions, requirements, risk severity, system configuration and accountable owner.

The evaluation contract freezes ordinary, edge and adverse cases with critical-case pass rules. Operational readiness supplies monitoring, incident escalation, rollback and backup/recovery evidence.

Before release, a named qualified reviewer examines unresolved failures and residual risk. The system records DENIED, HOLD, REJECTED, REQUEST_EVIDENCE or an authorized version-bound approval; no average metric overrides a failed critical permission or safety test.

**FAILED-GATE TRANSITION - MISSION-CRITICAL FAIL SAFE**

When a gate fails during operation, the orchestrator first prevents publication and side effects, preserves input and run lineage, quarantines dependent outputs, alerts the owner, investigates permission/source/model state, and creates a corrected revision.

The affected validations rerun; reviewer authority and the exact revision are checked again. A restored backup must reproduce the old state without applying current rules retroactively.

**Validation case matrix**

| Case | Required state | Release rule |
|---|---|---|
| Normal | Expected bounded candidate and explicit release checks | Pass only against frozen expected outcome. |
| Missing source | Evidence gap / request evidence | No unsupported critical field may be released. |
| Contradictory source | Conflict state / HOLD | Do not collapse disagreement into a settled claim. |
| Inaccessible source | Source-health failure | Do not report no-change. |
| Unauthorized retrieval | DENIED / no semantic match | No prohibited content may be laundered through synthesis. |
| Stale review | Reject prior approval | Recheck authority against exact candidate revision. |
| Unsafe tool | Quarantine / no side effect | Write action remains separately scoped and denied. |

## 7. PILOT MEASUREMENT, METRICS & MATURITY

A narrow pilot tests one authorized workflow with a declared baseline. Measure authoritative-version identification time, change-to-reviewed-impact time, material-field grounding, reconstructable-run coverage, decision completeness, correction and recovery, plus net operational burden.

Prespecify stop conditions for authority ambiguity, approval bypass, unauthorized disclosure, silent source failure or unreviewable critical output. A failed pilot narrows, redesigns or stops the use case.

Architecture evidence supports architectural reasoning. Implementation requires a controlled repository and executable artifact. Performance requires frozen, task-specific tests and failure analysis.

Security requires an assessment and remediation record. A customer outcome requires an authorized pilot and observed baseline. Validated use requires organization-specific qualification and approval. No movement between these states follows from wording in a whitepaper.

### 7.1 Pilot acceptance and stop conditions

An authority/source gate requires an approved source, purpose and reconstructable version; ambiguity stops release. Candidate quality requires grounded material fields and explicit unknown/conflict states; unsupported critical fields stop the workflow. Human authority requires approve, revise, request-evidence and reject operations with retained rationale; approval bypass stops the pilot. Security and privacy require negative access tests and absence of restricted disclosure; a critical exposure stops use. Reliability requires visible retry, recovery and terminal states; silent failure or irrecoverability stops expansion. Value requires a prespecified baseline and observation period, including review and control burden; an unmeasured benefit is not a pilot success.

### 7.2 Evidence maturity classes

The public source supports the architecture class. Runnable implementation, functional performance, security posture, customer adoption, commercial value and organization-specific validated use are separate classes with separate evidence requirements. An independent design review can strengthen architecture confidence but does not promote the system to production. A repository, frozen test set and observed browser/API run would support implementation and narrow functional claims. An authorized pilot plus observed baseline would support context-specific operational claims; qualified validation remains the deploying organization's responsibility.

**PILOT MEASUREMENT MATRIX**

| Dimension | Required denominator / evidence | Stop condition |
|---|---|---|
| Authority/source | Approved source + purpose + reconstructable version | Authority ambiguity |
| Candidate quality | Grounded material fields + explicit unknown/conflict states | Unsupported critical field |
| Human authority | Approve / revise / request evidence / reject with rationale | Approval bypass |
| Security/privacy | Negative access tests + absence of restricted disclosure | Critical exposure |
| Reliability | Visible retry, recovery and terminal states | Silent failure or irrecoverability |
| Value | Prespecified baseline + observation period + review/control burden | Unmeasured benefit |

**Evidence maturity classes**

| Class | Evidence required | Claim boundary |
|---|---|---|
| Architecture | Public source / design rationale | Supports architectural reasoning only. |
| Implementation | Controlled repository + executable artifact | Supports runnable implementation claims. |
| Functional performance | Frozen task-specific tests + failure analysis | Supports narrow measured performance claims. |
| Security posture | Assessment + remediation record | Supports assessed security posture. |
| Customer outcome | Authorized pilot + observed baseline | Supports context-specific operational outcomes. |
| Validated use | Organization-specific qualification + approval | Supports validated use in the deploying organization. |

## 8. LIMITATIONS AND CONCLUSION

Source coverage, parsing, retrieval, professional applicability, reviewer consistency and model behavior remain context dependent. Rare failures may escape a finite evaluation. Enterprise cost and benefit have not been measured for TRACE. Its falsifiable proposition is that attaching authority and evidence through candidate, validation, decision and outcome makes the workflow more reviewable and correctable. Production value must be demonstrated in a bounded authorized pilot.

## 9. ADVERSE SCENARIOS AND REQUIRED STATES

An official source becomes unavailable during a scheduled check: capture the attempted retrieval and source-health failure, preserve the last known baseline as historical evidence and do not report 'no change.' A document is replaced by a page with the same title but a different lifecycle status: retain both snapshots and put the affected impact candidate on HOLD. A model cites a valid document but the wrong passage: the citation check fails the material claim. A user loses reviewer authority while a case is paused: recheck authorization at disposition and reject the old action. A source changes after approval: calculate dependent candidates, retain prior history and require revision-bound review. Each scenario needs expected/actual evidence at the API and UI layers.

**ADVERSE SCENARIO CONTROL MATRIX**

| Scenario | Required evidence/state | Control outcome |
|---|---|---|
| Official source unavailable | Capture attempted retrieval + source-health failure; preserve last known baseline as historical evidence. | Do not report no change. |
| Same title, changed lifecycle status | Retain both snapshots; compute affected impact candidate. | HOLD. |
| Valid document, wrong cited passage | Resolve citation to stored version and relevant location. | Fail the material claim. |
| Reviewer authority revoked while paused | Recheck authorization at disposition. | Reject the old action. |
| Source changes after approval | Calculate dependent candidates; retain prior history. | Require revision-bound review. |

Every adverse scenario requires expected and actual evidence at both API and UI layers.

## 10. OBSERVABILITY AND RECOVERY CONTRACT

Monitor connector success, baseline age, parsing gaps, candidate queues, review latency, denial counts, failed grounding, retries, terminal states and restore success. Each metric needs an observation window, denominator and ownership.

An audit event must identify input/source versions, workflow revision, principal, operation, decision and outcome without leaking secrets or sensitive content into logs. Recovery testing reconstructs a historical decision from archived originals and configuration, not from today's source snapshot. Backup, rollback and incident response remain organization-specific operational controls.

| Plane | Telemetry | Rule |
|---|---|---|
| Connector / source | Connector success; baseline age; parsing gaps | Observation window + denominator + owner |
| Workflow | Candidate queues; review latency; retries; terminal states | Visible failure and recovery states |
| Assurance | Denial counts; failed grounding | No silent coercion of failed values |
| Recovery | Restore success | Reconstruct historical decision from archived originals/configuration |

## 11. RESEARCH AGENDA AND FALSIFICATION

Compare the complete evidence path with the deploying team's current workflow on identical authorized cases. Freeze the answer key, source cutoff and reviewer rubric before observing outputs. Record missed material changes, unsupported claims, wrong applicability, reviewer correction time, rework and incident burden.

A plausible failure of TRACE is that additional governance overhead outweighs useful preparation savings; another is that incomplete source coverage makes review less reliable despite a polished interface. The pilot must expose those outcomes. No architecture diagram can settle a question that requires measured task performance and qualified review.

**FALSIFICATION BASELINE CONTROLS**

| Control | Required baseline |
|---|---|
| Comparison population | Identical authorized cases in TRACE and current workflow |
| Frozen before observation | Answer key; source cutoff; reviewer rubric |
| Error outcomes | Missed material changes; unsupported claims; wrong applicability |
| Operational burden | Reviewer correction time; rework; incident burden |
| Failure hypothesis 1 | Governance overhead outweighs useful preparation savings |
| Failure hypothesis 2 | Incomplete source coverage reduces review reliability despite a polished interface |
| Decision principle | Measured task performance and qualified review, not architecture diagrams, determine the result |

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PRAMANEX TRACE is presented as a reference architecture and operating contract. Production deployment, customer outcome, independent security assessment and organization-specific validation are not established by this paper.